

DRISHYAM

From Footage to Action

Product Requirements Document (PRD)

Real-time AI Surveillance Platform

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1. Product Overview

Drishyam is a real-time AI-powered video surveillance platform that converts passive CCTV cameras into an active security system. It continuously analyzes live video streams, detects threats (weapons, violence, intrusion, suspicious behavior), triggers immediate alerts, and enables rapid response — moving from “footage” to “action” within seconds.

2. Problem Statement

Traditional CCTV systems only record events. Human operators suffer attention decay after ~20 minutes. Motion-based alerts generate 90–98% false alarms. By the time a real threat is noticed, the incident is often over. Security teams waste resources, and preventable crimes continue under cameras.

3. Target Users & Personas

Primary:

- Facility / Security Managers (mid-market commercial, industrial, warehouses, schools, hospitals)
- Security Operators / Control Room Staff
- Organization Admins (multi-site businesses)

Secondary:

- Business Owners / CFOs (ROI & cost control)
- System Integrators / Installers
- Government / Smart City operators (later phase)

4. Core Value Proposition

- Detect real threats in < 2–5 seconds on live streams
- Reduce false alarms by 80–95% compared to motion detection
- Deliver verified alerts with evidence clips to the right people instantly
- Work with existing IP cameras (camera-agnostic where possible)
- Privacy-first design (edge processing options, face blur, DPDP alignment)

5. Feature List (Prioritized)

5.1 MVP (Must Have)

- Live RTSP/ONVIF camera ingestion (multi-camera)
- Real-time AI detection: Weapon (gun/knife), Violence/Fight, Intrusion / Zone breach, Loitering
- Confidence scoring + false-positive filtering
- Instant alert generation with short evidence clip (5–15 sec)
- Push notification + in-app alert feed (Web + Mobile)
- Alert acknowledgment + basic status workflow (New → Acknowledged → Resolved)
- Simple multi-tenant organization + site + camera hierarchy
- Basic dashboard: live grid, alert list, camera health
- User roles: Org Admin, Site Manager, Operator, Viewer
- Privacy blur option for faces in stored clips
- Audit log of alerts and user actions

5.2 Phase 2 (Should Have)

- Facial recognition (blacklist / VIP) with explicit consent & compliance controls
- Crowd density & traffic monitoring
- Natural language search on historical events
- Automated response actions (trigger alarm, lock door via IoT, PA announcement)
- Multi-site unified view & reporting
- Mobile app with live view + two-way audio
- Advanced analytics & heatmaps

5.3 Future

- SOS / Medical emergency detection
- Fire / smoke / environmental hazard detection
- Full IoT automation (lights, power cut, access control)
- Public announcement integration
- Predictive risk scoring

6. High-Level User Flows

Operator Alert Flow: Alert appears → Operator opens clip → Reviews context → Acknowledges / Escalates / Marks False Positive → System learns (optional feedback loop)

Admin Setup Flow: Create Org → Add Site → Add Cameras (RTSP/ONVIF) → Configure Detection Zones & Rules → Invite Users → Go Live

Incident Response Flow: Threat detected → Alert + Clip generated → Notified users → Optional automated action → Evidence stored → Report generated

7. Business Rules

- Only users belonging to an organization can see that organization's cameras and alerts.
- Alerts are scoped to the site/camera the user has permission for.
- False-positive feedback can be used to improve model thresholds (opt-in).
- Clips older than configured retention period are auto-deleted (default 30 days, configurable).
- High-severity alerts (weapon, violence) cannot be auto-dismissed.
- All sensitive actions (user invite, camera delete, role change) are audit-logged.
- Edge inference is preferred; cloud fallback only when edge is unavailable or for heavy models.

8. MVP Scope (In / Out)

IN: Real-time detection of 4 core classes, alerting, evidence clips, basic multi-tenant dashboard, roles, privacy blur, RTSP ingestion.

OUT (MVP): Full facial recognition, complex IoT actuation, advanced search, white-label, offline-first mobile, public API marketplace, multi-language UI.

9. Success Metrics (MVP)

- Detection latency < 5 seconds (end-to-end)
- False positive rate < 10% on core classes in pilot environments
- Alert-to-acknowledgment median time < 60 seconds
- Pilot-to-paid conversion ≥ 40%
- System uptime ≥ 99.5% for ingestion + detection

10. Assumptions & Dependencies

- Customers have existing IP cameras with RTSP/ONVIF support
- Network bandwidth sufficient for at least low-resolution analysis streams
- Edge devices (or capable cameras) available for production deployments
- Team can fine-tune models on Indian-context video data